Tahmid Hasan Sakib

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PROFESSIONAL SUMMARY

Graduate Research Assistant applying Python/C++, Linux, NVIDIA Jetson, camera-LiDAR sensing, computer vision, and secure ROS 2 in scaled autonomous vehicle research. Experience validating embedded prototypes through controlled physical tests.

TECHNICAL SKILLS

Embedded software and platforms: Python, C/C++, Linux, Git, ROS 2/SROS2, NVIDIA Jetson TX2 and AGX Orin, Arduino/AVR, introductory Verilog; TensorRT in documented QCar2 work.

Perception and validation: YOLOv5n/YOLOv8, OpenCV, PyTorch, scikit-learn, Intel RealSense and CSI cameras, 2D LiDAR, CARLA, MATLAB, LTspice, Cisco Packet Tracer.

EXPERIENCE

Graduate Research Assistant | Tennessee Technological University, Cookeville, TN | May 2024-Present

- Validated Jetson TX2 camera-LiDAR perception and three-stop QCar1 control in clean and printed-sign physical runs.
- Recorded detector and runtime configurations for mitigation comparisons and analyzed missed-stop outcomes at vehicle level.
- Mapped SROS2 provisioning and authenticated DDS trust boundaries on a separate QCar2 platform; documented qualitative vehicle effects in controlled tests.
- Compared offline and live QCar2 monitoring/restoration evidence, tracking detector, flagging, and control outcomes with collaborators.
- Mentored undergraduate researchers in NSF REU cohorts in summers 2025 and 2026 under faculty supervision.

Hardware Engineer Intern | Robi Axiata Limited, Dhaka, Bangladesh | December 2019-January 2020

- Prototyped AVR/Arduino systems and simulated hardware and communication behavior with MATLAB, LTspice, and Cisco Packet Tracer.

SELECTED PROJECTS

QCar1 physical adversarial perception | Jetson TX2, CSI camera, RPLiDAR A2, YOLOv5n

Runs with a missed stop changed from 8/18 to 0/18 across separate pre/post physical groups; detector and runtime settings both changed.

PULSE runtime anomaly monitoring | physical QCar2

Clean-calibrated monitoring flagged 18/20 patched-sign encounters in 10 live fallback-on runs; all 20 stop sequences completed, without an independent stop-line measure.

REDAP restoration-based defense | physical QCar2 frames

Offline analysis of 151 stored frames found 57/151 attacked versus 110/151 restored frames at or above the 0.60 confidence cutoff; later vehicle tests were separate.

Network Intrusion Detection for IoT Devices | GitHub repository

Independent Python notebook project covering CICIoT2023 binary, 8-class, and 34-class classification, feature selection, conventional ML, and ANN/LSTM/GRU.

EDUCATION

Tennessee Technological University | M.S., Electrical and Computer Engineering, expected December 2026 | GPA 4.00/4.00

Master's thesis defended; official degree conferral pending.

Military Institute of Science and Technology, Dhaka, Bangladesh | B.Sc., Computer Science and Engineering, August 2021

SELECTED PUBLICATIONS

Accepted journal article: First author, "Vulnerability Assessment of Adversarial Image Attacks in a Miniaturized Autonomous Vehicle Testbed without Relying on Way-Point Information," *IEEE Access*, accepted for publication, 2026.

Published conference papers: First author, SROS2 supply-chain study, IEEE MILCOM 2025; coauthor, physical naturalistic-patch study, IEEE SATC 2026.